ECE 411 Introduction to Machine Learning
Course Syllabus (Fall 2026)

Time & Location:

Lecture: MW 11:45 am to 1 pm, 1226 EB2.

Discussion: F12:50 to 1:40 pm, 1229 EB2.

Instructor: Dr. Chau-Wai Wong

Course webpage: https://ncsu-wong.org/ece411_26Fall/
[or Google "wong ncsu"]

Teaching Assistant: [See course webpage for most up-to-date information]

Course Description: Machine learning has progressed remarkably over the past decade and virtually every single industry has seen the encouraging potential brought by the application of machine learning. This course introduces fundamental concepts and algorithms in machine learning that are vital for understanding state-of-the-art and cutting-edge development in machine learning. This course exposes students to real-world applications via well-guided homework programming problems and a term project.

Recently Added Course Content:

1. Updated lecture content on large language models (LLMs) and diffusion models.
2. Newly introduced lab exercises (e.g., on using conda environment, HPC clusters, VS code remote development) and reduced homework load.
3. New project options on transformers, diffusion models, and federated learning.

Learning Outcomes: By the end of this course, the students should be able to:

1. Explain basic concepts and underlying working principles of machine learning algorithms such as linear and logistic regressions, maximum likelihood estimation, and expectation-maximization.
2. Derive and implement from scratch basic machine learning algorithms such as linear and logistic regressions.
3. Explain the pros and cons of common classical machine learning algorithms.
4. Identify the correct machine learning tool for solving a particular kind of problem.
5. Use off-the-shelf machine learning packages to address real-world machine learning problems, e.g., examples adapted from Kaggle competitions.
6. Gain an appreciation for ongoing advances in the field by explaining the key merits and application domains of a few exemplar cutting-edge algorithms like deep neural networks.

Prerequisites: (*i*) ST 300-level or above, and (*ii*) ECE301/CSC316/ISE361/MA341. Email the instructor your transcript to discuss if the second prerequisite can be waived.

Course Structure & Grading Components:

The course consists of two 75-min lectures and one 50-min discussion section per week. A teaching assistant will lead the discussion section, covering practice problems and answering questions from students. There will be weekly homework assignments (30%) that contain both written problems and programming problems, two midterm exams (20%×2), and one term project (30%). ***Failure to submit three (3) homework assignments without justification will result in a failing grade.*** Programming will be in Python or Matlab. Students are expected to be able to write computer programs and have mathematical maturity in probability theory (e.g., have taken a 300-level statistics course) before taking the course. A linear algebra course such as MA305/405 is recommended while taking the course.

Topics: *Classical topics:* regression, classification, support vector machines, and cross-validation. *Modern topics:* convolutional neural networks (CNN), long short-term memory (LSTM), transformers (e.g., BERT and GPT), and diffusion models.

Textbooks:

G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, Introduction to Statistical Learning with Applications in Python (ISLP), Springer, 2023. [Online]

T. Hastie, R. Tibshirani, J. Friedman, The Elements of Statistical Learning, (12th Printing, 2017), 2nd Ed., Springer. [Online]

I. Goodfellow, Y. Bengio, A. Courville, Deep Learning, MIT Press, 2016. [Online]

Reference Books:

G. James, D. Witten, T. Hastie and R. Tibshirani, Introduction to Statistical Learning with Applications in R, 1st edition, Springer, 2013. [Online]

S. H. Chan, Introduction to Probability for Data Science, 2021. [Online]

M. P. Deisenroth, A. A. Faisal, and C. S. Ong, Mathematics for Machine Learning, 2020. [Online]

J. L. Devore, Probability and Statistics for Engineering and the Sciences, Ed. 9.

A. Leon-Garcia, Probability, Statistics, and Random Processes for Electrical Engineering, Ed. 3, 2008.

C. E. McCulloch, S. R. Searle, Generalized, Linear, and Mixed Models, Wiley & Sons, 2001.

K. P. Murphy, Machine Learning: A Probabilistic Perspective, 2012. [Online via NC State library]

R. O. Duda, P. E. Hart, D. G. Stork, Pattern Classification, 2nd Ed., Wiley, 2001.

E. Alpaydin, Introduction to Machine Learning, 2nd Ed., MIT Press, 2009.

Letter Grades:

This Course uses Standard NC State Letter Grading:

97	≤	**A+**	≤	100
93	≤	**A**	<	97
90	≤	**A-**	<	93
87	≤	**B+**	<	90
83	≤	**B**	<	87

80	≤	**B-**	<	83
77	≤	**C+**	<	80
73	≤	**C**	<	77
70	≤	**C-**	<	73
67	≤	**D+**	<	70
63	≤	**D**	<	67
60	≤	**D-**	<	63
0	≤	**F**	<	60

Requirements for Credit-Only (S/U) Grading: To receive a grade of S, students are required to take all exams, complete all assignments, and earn a grade of C- or better. Conversion from letter grading to credit only (S/U) grading is subject to university deadlines. Refer to the Registration and Records calendar for deadlines related to grading. For more details refer to http://policies.ncsu.edu/regulation/reg-02-20-15.

Auditing Policy: Students must register this course for auditing. Auditing students must turn in all homework assignments. However, they are not required to take the exams. More information about and requirements for auditing a course can be found at

http://policies.ncsu.edu/regulation/reg-02-20-04

Homework Assignment Policy:

Due date & time: All assignments must be turned in before the beginning of class at 11:45 am on the date they are due. The exact date will be indicated on the course webpage. If there is a discrepancy between the due dates listed on the course webpage and Gradescope, you must follow the due date indicated on the course webpage.

Final homework grade: Failure to submit three (3) homework assignments without justification will result in a failing grade. ***The lowest homework grade will be dropped*** at the end of the semester, so you have one “free pass.” Please use it sparingly, e.g., to allow for the possibility of illness of yourself, unanticipated events/incidents about your family, and short-noticed deadlines of other courses. You are expected to turn in your assignments on time for any anticipated conflicts, so please plan accordingly.

Late assignment: You may turn in any assignment ***one day late for up to 75% credit*** and ***no other extensions will be given***. There will be no make-up assignments even for those who have valid excuses to miss an assignment per the university regulation. The remedy for unexpected situations is built into the policy of dropping the lowest homework grade. There will be no response to requests for extensions on assignments and excused assignments.

Submission: Homework will be submitted electronically via Gradescope. You will be required to mark the pages corresponding to different problems and subproblems when you make an upload. You may always upload multiple times if you make a mistake in marking. You will receive a zero for any problems not corrected identified/marked.

Regrade policy: All requests for regrades must be made through Gradescope ***within 7 days***. Other than regrade requests, no verbal complaints or email complaints will be considered. Before submitting any request for partial credit, please keep in mind that the first objective of grading is to be consistent. It may seem unfair that you did not get as much partial credit as you think you deserve. Keep in mind, however, that this may have been consistently applied to all students thus no more partial credit may be given. Mistakes can be made in the grading process, which the TAs will be happy to correct, but it is unlikely that more partial credits will be given.

Late/Missed Exams:

Emergency or unanticipated absences will be handled on a case-by-case basis. There will be no make-up exam. Grades for a missing item with valid excuses and documentation will be estimated using the grades of other available items in the same category. Students who believe they have valid excuses to miss an exam must comply with University Attendance Regulations REG 02.20.03 on NC State's website.

Attendance Policy: Students must attend the class in person. For a detailed Attendance/Absence Policy, see Attendance Regulation NCSU REG 02.20.03 at http://policies.ncsu.edu/regulation/reg-02-20-03.

Absences Policy: Excuses for unanticipated absences must be presented to the instructor within one week after the return to class.

Academic Integrity:

Discussion of homework questions is encouraged but please submit your own independent homework solutions.

Students are required to comply with the university policy on academic integrity found in the Code of Student Conduct found at http://policies.ncsu.edu/policy/pol-11-35-01

Violations of academic integrity will be handled per the Student Discipline Procedures (NCSU REG 11.35.02).

Honor Pledge: Your signature on any test or assignment indicates "I have neither given nor received unauthorized aid on this test or assignment."

Digital Course Components: Students may be required to disclose personally identifiable information to other students in the course, via digital tools, such as email or web postings, where relevant to the course. Examples include online discussions of class topics and posting of student coursework. All students are expected to respect the privacy of each other by not sharing or using such information outside the course.

Course webpage (course portal): [Google "ncsu wong", find Dr. Wong's webpage, and go to the "Teaching" tab]

Piazza (for Q&A) [for hyperlink, see course webpage]

Gradescope (for homework submission) [for hyperlink, see course webpage]

Accommodations for Disabilities:

Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with the Disability Resource Office at Holmes Hall, Suite 304, Campus Box 7509, 919-515-7653. For more information on NC State’s policy on working with students with disabilities, please see the Academic Accommodations for Students with Disabilities Regulation (REG02.20.01) (https://policies.ncsu.edu/regulation/reg-02-20-01/).

Accommodations for Mental Health Concerns and Stressful Events:

As a student, you may experience a range of personal issues that can impede learning, such as strained relationships, increased anxiety, alcohol/drug concerns, feeling down, difficulty concentrating and/or lack of motivation. These mental health concerns or stressful events may lead to diminished academic performance and may impact your ability to participate in daily activities. It is very important that you have a support system and that you ask for help when you are struggling. The Counseling Center at NC State offers confidential mental health services for full time NC State students, including ***same-day emergency services***. Please visit https://counseling.dasa.ncsu.edu/ to get connected.

The WolfPack Wellness Resources (https://wellness.ncsu.edu/resources/) is available to everyone. At the bottom of link, you can find the Mental Health First Aid (MHFA) and QPR Training programs.

Non-Discrimination Policy:

NC State provides equal opportunity and affirmative action efforts, and prohibits all forms of unlawful discrimination, harassment, and retaliation ("Prohibited Conduct") that are based upon a person’s race, color, religion, sex (including pregnancy), national origin, age (40 or older), disability, gender identity, genetic information, sexual orientation, or veteran status (individually and collectively, “Protected Status”). Additional information as to each Protected Status is included in NCSU REG 04.25.02 (Discrimination, Harassment and Retaliation Complaint Procedure). NC State’s policies and regulations covering discrimination, harassment, and retaliation may be accessed at http://policies.ncsu.edu/policy/pol-04-25-05 or https://oied.ncsu.edu/divweb/. Any person who feels that he or she has been the subject of prohibited discrimination, harassment, or retaliation should contact the Office for Equal Opportunity (OEO) at 919-515-3148.

Course Evaluation: Students will be notified during the last few weeks of the class to complete course evaluations. All evaluations are confidential; instructors will never know how any one student responded to any question, and students will never know the ratings for any particular instructors.